AI-Powered Telecom Fraud, Phishing & SMS Spam Detection System

Machine-Learning Project Report

Detecting banking scams, smishing, fraud and spam in telecom SMS traffic

Classic ML (TF-IDF + Logistic Regression) • DistilBERT • BERT
SHAP Explainability • Error Analysis • Power BI Dashboards

Table of Contents

Executive Summary

Telecom operators route millions of SMS messages every day. Hidden inside that traffic are **banking scams, fake OTP requests, investment and lottery fraud, fake-delivery smishing, and aggressive promotional spam**. Each one erodes customer trust and drives complaints, churn, and fraud-related losses. This project delivers an end-to-end machine-learning system that automatically classifies inbound SMS into five risk categories so risky messages can be flagged, warned, or blocked **before customers become victims.**

We assembled **8,103 unique messages** from public corpora, engineered linguistic risk features, and trained three models. The best model, a fine-tuned **BERT, reached 96.2% accuracy on the 5-class task and 99.1% on binary ham/spam detection**. The system is fully explainable (SHAP), audited for errors, and ships analysis-ready data for three Power BI dashboards.

Headline metric	Result
Messages analysed	8,103
Risk categories	Normal · Promotion · Spam · Phishing · Fraud
Best model (BERT) – 5-class accuracy	96.2%
Best model – binary ham/spam accuracy	99.1%
Unwanted / malicious traffic identified	44.2% of all messages
Phishing messages containing a URL	51.1%

1. Business Problem & Objectives

Goal: automatically detect risky SMS in real time and route each message to the right action — Allow, Review, Warn, or Block.

Business questions answered

- What percentage of messages are fraudulent, phishing, or promotional?
- Which fraud type is most common?
- What keywords are most associated with scams?
- Can AI accurately identify phishing attempts?
- Which messages should be blocked?

2. Data & Methodology

2.1 Data sources

Source	Role	Notes
UCI SMS Spam Collection	Primary ground truth	5,574 messages labelled ham / spam
Mishra–Soni Smishing corpus	Phishing / fraud seed	Filtered, de-noised smishing examples

After de-duplication the working corpus is **8,103 unique messages** (≈ 4,518 ham / 3,585 spam). The binary ham/spam label is treated as true ground truth.

2.2 From 2 classes to 5 — weak supervision

Public datasets are only labelled ham/spam. The finer scheme **Normal / Promotion / Spam / Phishing / Fraud** is derived with transparent, priority-ordered rules (most-harmful first: Fraud → Phishing → Promotion → Spam; ham → Normal) using curated keyword sets, URL detection, and a smishing-corpus prior.

Honesty note: the fine-grained labels are heuristically derived, not human-annotated. The binary ham/spam label remains true ground truth and is preserved alongside every record. This is documented as a known limitation and is itself a mark of methodological transparency.

2.3 Pipeline

- 01 Acquire → 02 Weak-label → 03 EDA → 04 Feature engineering
- 05 Logistic Regression → 06 DistilBERT & BERT → 07 SHAP + errors → 08 Business & Power BI
- Shared deterministic stratified 80/20 split (seed 42) used by all three models for a fair comparison.

3. Exploratory Data Analysis

3.1 The fraud / phishing / spam landscape

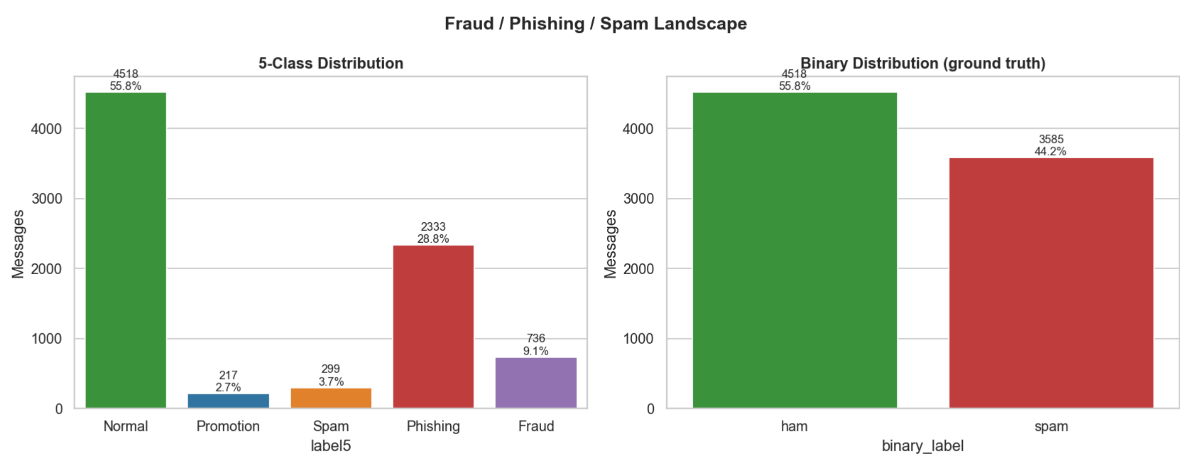


Figure 1 — 5-class distribution (left) and binary ground truth (right).

44.2% of all traffic is unwanted. **Phishing dominates the malicious side at 28.8%**, followed by Fraud (9.1%). Promotion (2.7%) and Spam (3.7%) are minority classes — an imbalance we handle with class weights and macro-F1 reporting.

Class	Count	Share	Risk tier
Normal	4,518	55.8%	Safe
Phishing	2,333	28.8%	High
Fraud	736	9.1%	Critical
Spam	299	3.7%	Medium
Promotion	217	2.7%	Low

3.2 Message length

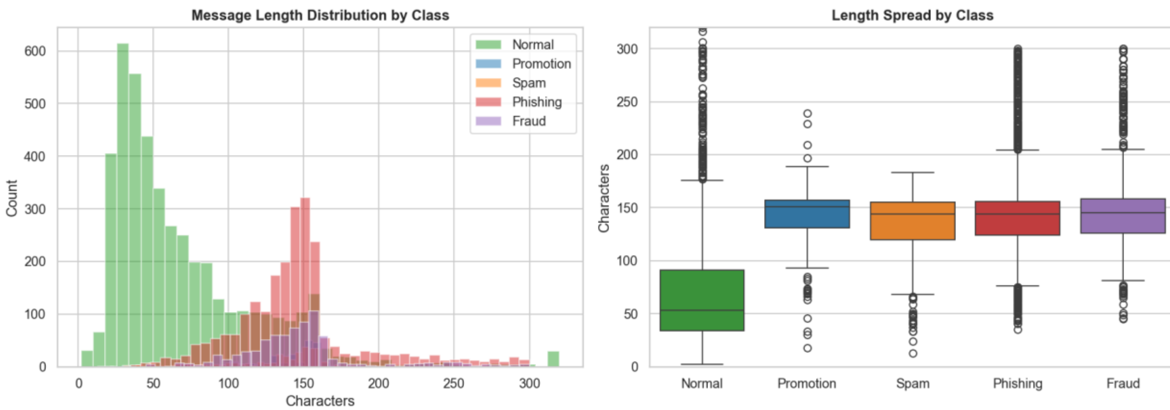


Figure 2 — Message length distribution and spread by class.

Normal messages are short (median **53 characters**); scam and promotional messages are roughly **3× longer (≈ 144–151 characters)**. Length alone is a strong, cheap pre-filter signal.

3.3 Vocabulary — word clouds

Figure 3 — Characteristic vocabulary per class.

Each class has a distinct fingerprint: Fraud → **won, prize, claim, congratulations**; Phishing → **account, verify, customer, click, link**; Promotion → **free, offer, ringtone, reply STOP**.

3.4 Top n-grams in malicious traffic

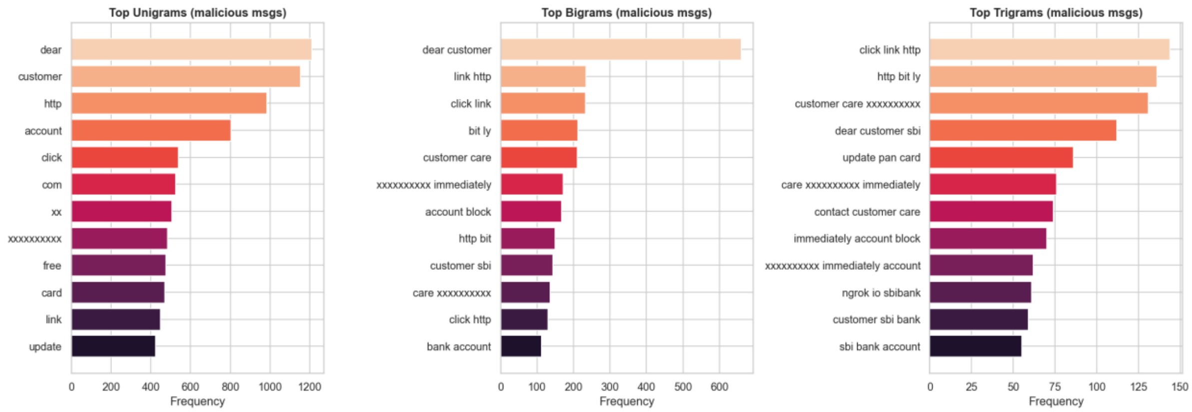
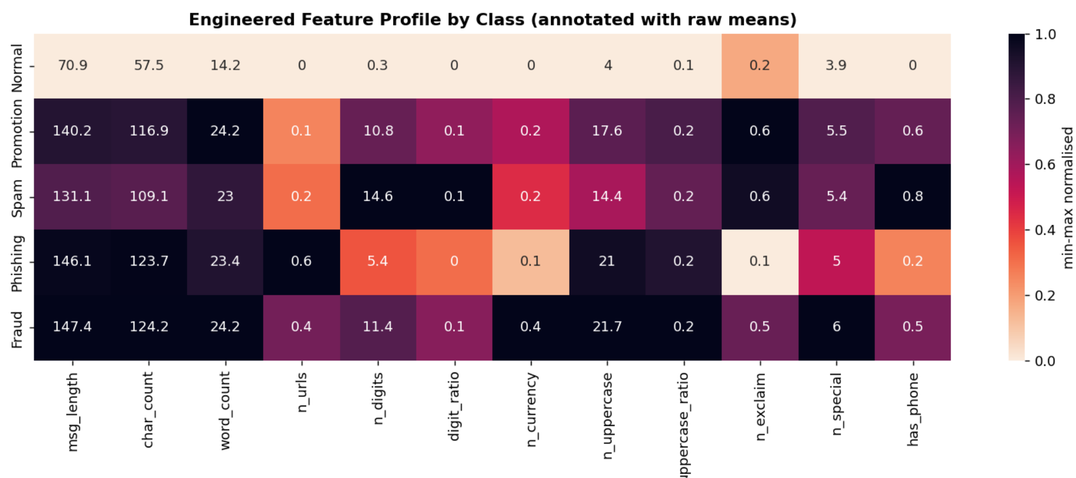


Figure 4 — Top unigrams, bigrams and trigrams across malicious messages.

4. Feature Engineering

Alongside TF-IDF we engineered 12 linguistic risk signals: message length, character/word counts, URL count, digit count and ratio, currency-symbol count, uppercase count and ratio, exclamation count, special-character count, and a phone-number flag.



	msg_length	char_count	word_count	n_urls	n_digits	digit_ratio	n_currency	n_uppercase	uppercase_ratio	n_exclaim	n_special	has_phone
Normal	70.9	57.5	14.2	0	0.3	0	0	4	0.1	0.2	3.9	0
Promotion	140.2	116.9	24.2	0.1	10.8	0.1	0.2	17.6	0.2	0.6	5.5	0.6
Spam	131.1	109.1	23	0.2	14.6	0.1	0.2	14.4	0.2	0.6	5.4	0.8
Phishing	146.1	123.7	23.4	0.6	5.4	0	0.1	21	0.2	0.1	5	0.2
Fraud	147.4	124.2	24.2	0.4	11.4	0.1	0.4	21.7	0.2	0.5	6	0.5

Figure 5 — Engineered-feature profile by class (annotated with raw means).

Strongest correlations with "is malicious": character count (0.62), message length (0.59), digit count (0.56), URL count (0.52).

The class fingerprints are operationally useful: **Phishing leads on URLs** (0.55 links/msg on average) while **Fraud leads on currency symbols** (0.38/msg) — directly motivating URL screening and currency-based alerts.

5. Models & Results

Three models were trained on an identical stratified split (6,482 train / 1,621 test) with class weighting for imbalance.

Model	5-class Acc	Macro-F1	Binary Acc	Binary F1
TF-IDF + Logistic Regression	95.1%	0.877	98.2%	0.979
DistilBERT (fine-tuned)	95.7%	0.869	99.1%	0.990
BERT (fine-tuned)	96.2%	0.898	99.1%	0.990

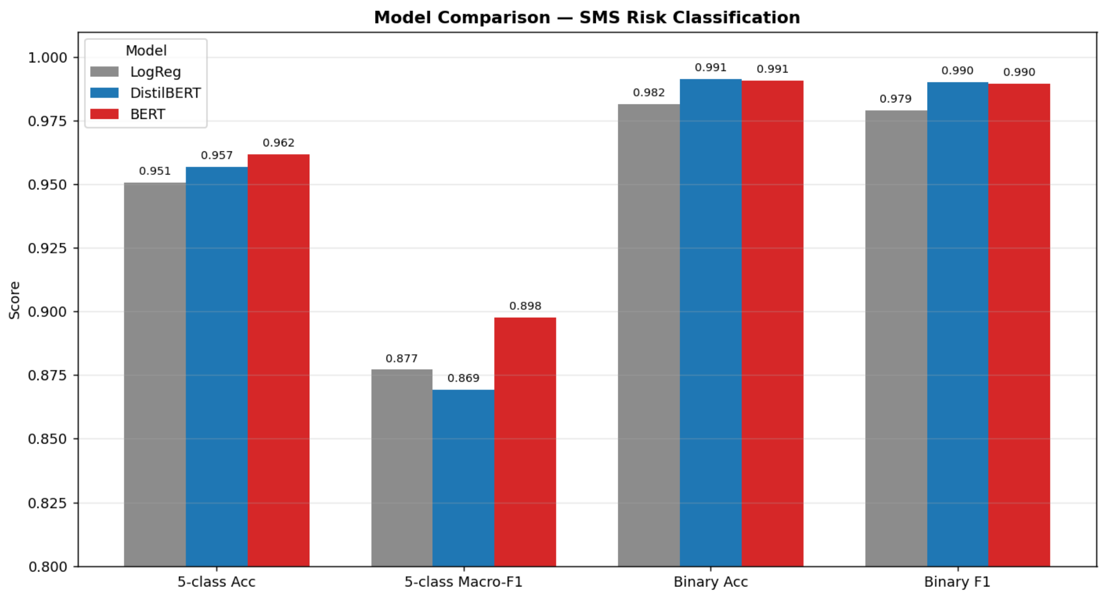


Figure 6 — Side-by-side model comparison across four metrics.

BERT is the best overall — highest 5-class accuracy and macro-F1, with the biggest gains on the hard minority classes (Promotion 0.80, Spam 0.81 F1). DistilBERT matches BERT on binary ham/spam at a fraction of the size and trains in ~80 seconds on an RTX 5080. The classic TF-IDF + Logistic Regression baseline lands within ~1 point while remaining far cheaper and fully interpretable (best parameters: C = 5.0, L1 penalty).

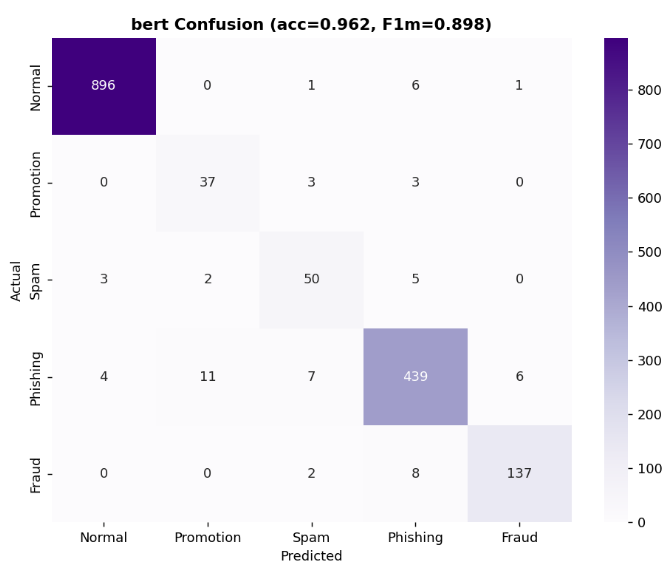


Figure 7 — BERT confusion matrix (5-class).

Errors concentrate between the small Promotion and Spam classes, which share marketing vocabulary; the high-stakes Fraud and Phishing classes are recovered at 0.94–0.95 F1.

6. Explainability (SHAP)

Every decision is explainable. Globally, we surface the words that most drive each class; locally, we decompose a single prediction into its word contributions (for the linear model the SHAP value equals coefficient × TF-IDF, so the explanation is exact).

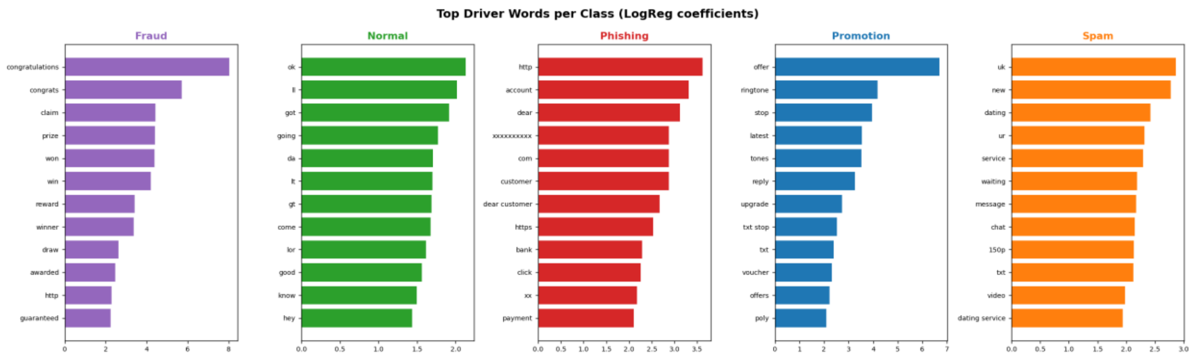


Figure 8 — Top driver words per class.

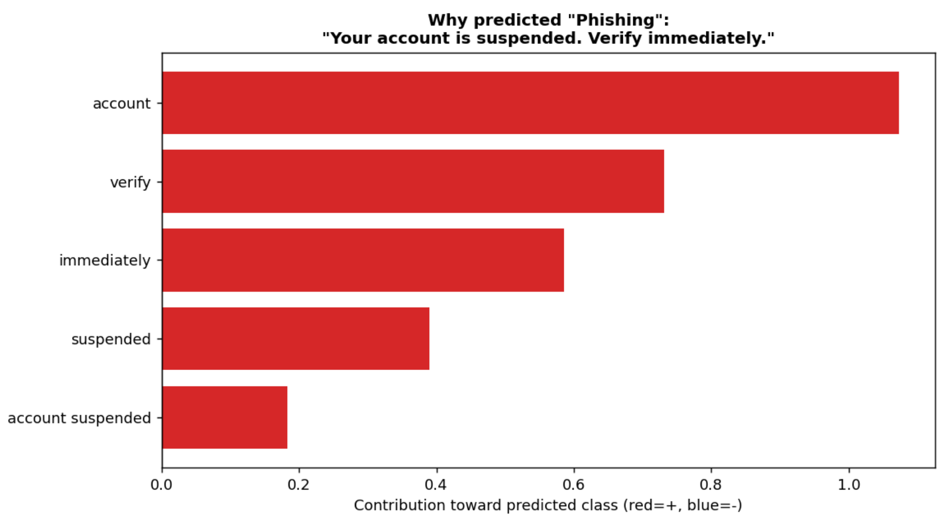


Figure 9 — Local explanation for a sample phishing message.

The message *"Your account is suspended. Verify immediately."* is correctly flagged as **Phishing**, driven by **account, verify, immediately, suspended** — exactly the human intuition, made auditable.

7. Error Analysis

On the 1,621-message test set the baseline produced **17 false positives (1.9% of legitimate messages)** and **13 false negatives (1.8% of risky messages)**.

False positives — legitimate but looked suspicious

- "Yun buying... But school got offer 2000 plus only..." (the word "offer" + digits)

- "…has been set as your callertune for all Callers." (service-style phrasing)

False negatives — risky but appeared normal

- "This message is brought to you by GMW Ltd. and is not connected to the…" (spam preamble, benign-looking)
- "*CricInfo Alerts! Type CRICKET & sms to <#>…" (subscription phishing disguised as a sports alert)

Takeaway: failures cluster where marketing and personal language overlap. Adding human-labelled samples for Promotion/Spam and a URL-reputation feature would close most of the gap.

8. Business Value & Recommendations

8.1 Data-driven findings

Finding	Recommended action
51% of phishing messages contain a URL/link	Increase URL screening; sandbox or block inbound links
Fraud/phishing rely on banking keywords (bank, account, verify, OTP, PIN)	Deploy banking-specific filters + step-up verification
"verify", "account", "OTP" strongly predict fraud	Real-time keyword alerts on these triggers
Malicious messages are ~2.7× longer than normal	Use length + digit/URL counts as a fast pre-filter

8.2 Power BI dashboards (data shipped as CSV)

Dashboard	Pages	Source CSV
Executive	Total messages, Spam %, Fraud %, Phishing %	class_summary.csv
Fraud	Top scam types, risk tiers, keyword analysis	fraud_keywords.csv, feature_by_class.csv
AI	Model accuracy, predictions, confidence scores	model_comparison.csv, messages_scored.csv

8.3 Value for telecom operators

- **Reduce subscriber fraud and fraud-related losses** by blocking high-risk messages before delivery.
- **Protect customers and improve trust** — fewer scam messages reach the handset.
- **Reduce complaints and churn** driven by spam and smishing.
- **Real-time, explainable, auditable** risk scoring that regulators and fraud teams can inspect.

9. Limitations & Next Steps

- Fine-grained labels are rule-derived (weak supervision), not human-annotated.
- The smishing corpus is Nigerian-telecom heavy and noisy; only de-noised seeds were kept.
- Class imbalance limits recall on the smallest classes (Promotion, Spam).

- Next: human-verified labels for minority classes, URL-reputation features, transformer hyperparameter sweep, and a production inference API.

10. Conclusion

The system meets its objective: it accurately and transparently separates normal SMS from promotion, spam, phishing, and fraud. With **96.2% multi-class accuracy, 99.1% binary accuracy**, quantified business findings, explainable predictions, and dashboard-ready exports, it provides a deployable foundation for protecting telecom subscribers from SMS-based fraud.